
(ECE 3113) Energy Conversion
Mid-Term Exam – Fall 2023
(40 minutes)

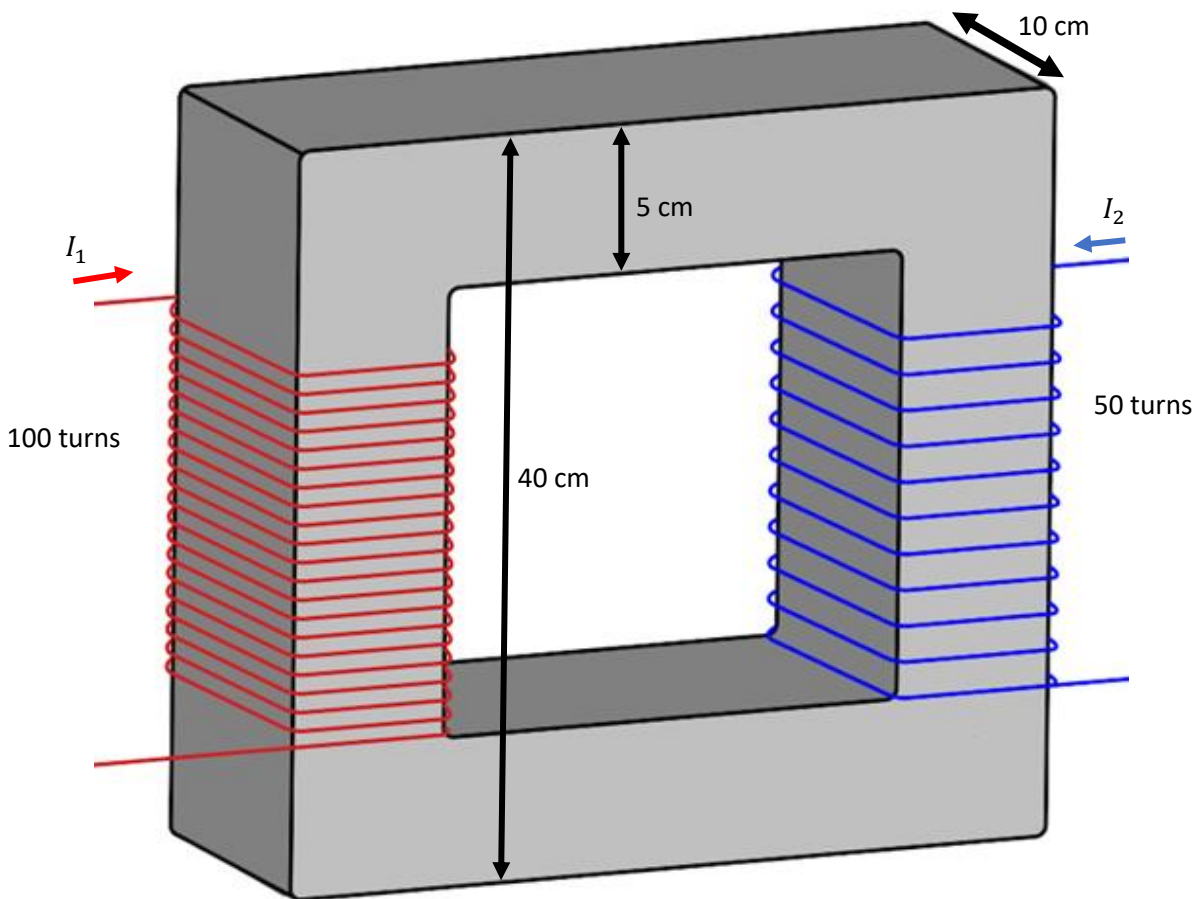
- Make sure that you write in your OU ID # and name.
 - Your answers are required to be shown to three significant digits.
 - Please show all your work to make it clear how you arrived at your solution.
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Problem 1 (10 points)

Provide a detailed derivation to show that the sum of currents at the neutral point in a 3-phase circuit with a positive sequence, where the instantaneous current for phase a is $i_a(t) = i_m \sin(2\pi 60t - 30)$, would be zero. Please include two major steps and the final expression.

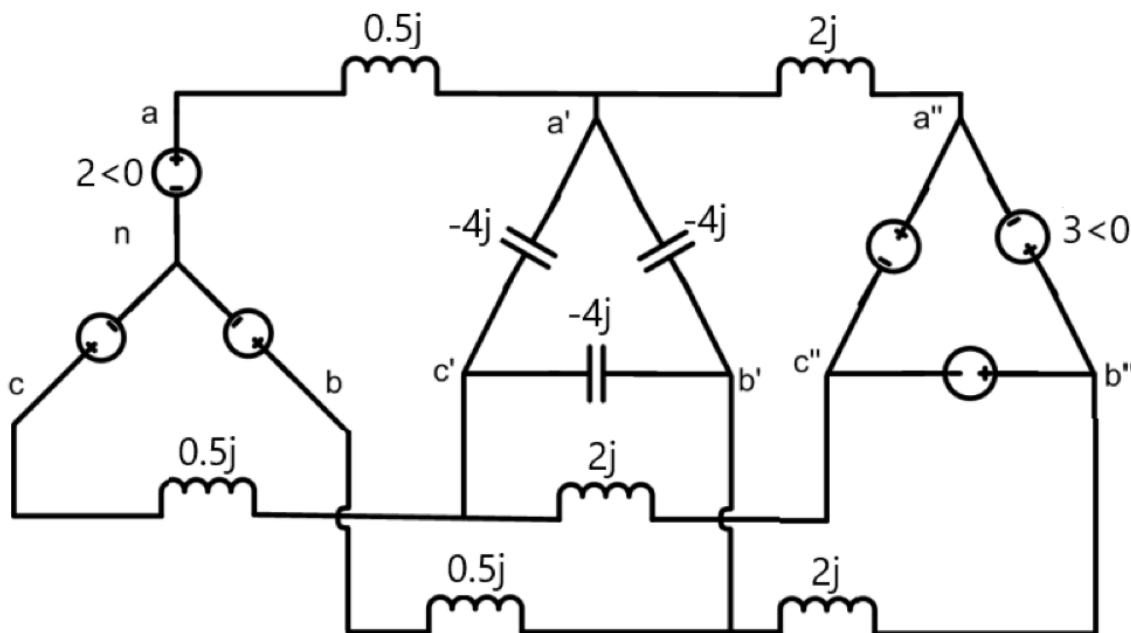
Problem 2 (30 points)

Consider the two-legged core shown below. The winding on the left leg of the core (N_1) has 100 turns, and the winding on the right (N_2) has 50 turns. The coils are wound in the directions shown in the figure. If each leg is 40 cm long (horizontal and vertical legs have same dimensions) and 5 cm wide, then calculate the flux produced by currents $i_1 = 2\text{ A}$ and $i_2 = 4\text{ A}$? (Assume $\mu_r \mu_0 = 12.56 \times 10^{-4}$ and constant) (Core depth = 10 cm).



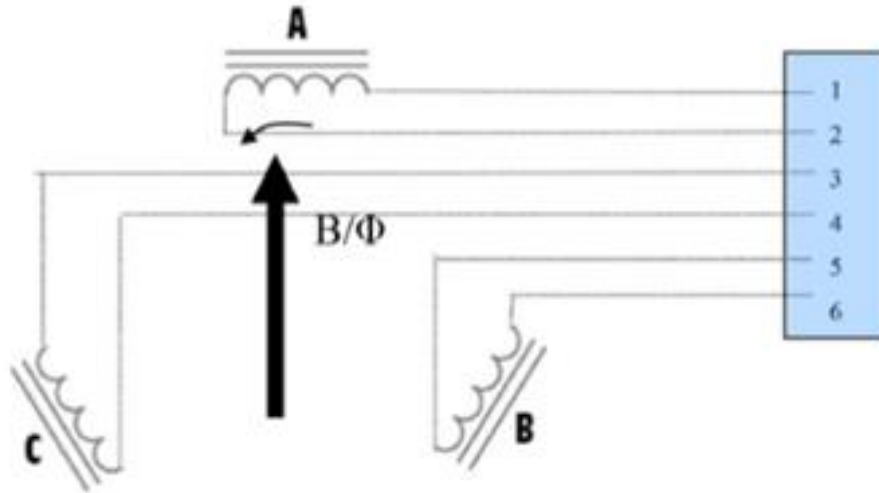
Problem 3 (30 points)

Assume a 3 ϕ system, Y-connected generator with $V_{an} = 2\angle 0^\circ$ volts supply a Δ -connected load with $Z_{\Delta} = -4j\Omega$ through a transmission line with impedance of $0.5j\Omega$ per phase. The load is also connected to a Δ -connected generator with $V_{a''b''} = 3\angle 0^\circ$ through a second transmission line which has an impedance of $2j\Omega$ per phase. 1) write the conversion equations (10points), 2) calculate the equivalent impedance (10points), and 3) draw the per phase representation of the given 3 phase circuit (10points).



Problem 4 (10 points)

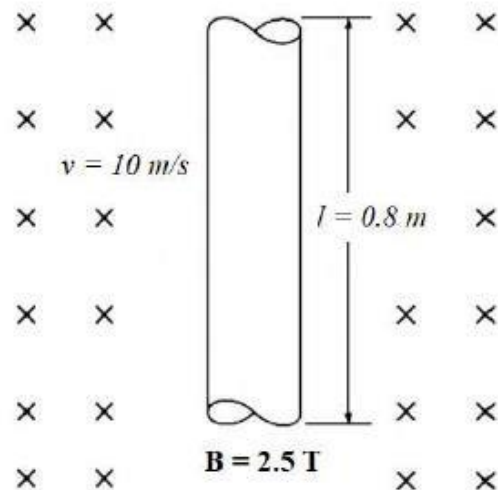
In the balanced 3 ϕ system shown, make a Delta-connection of this balanced three phase circuit at blue box at the terminal.



Problem 5 (20 points)

Consider that the wire of length $l = 0.8 \text{ m}$, shown below,

- a) is moving in the presence of a magnetic field. Find the induced voltage in the wire; when $v = 10 \text{ m/s}$, $B = 2.5 \text{ T}$ and the moving direction is perpendicular to the wire, as shown. Please write the equation and final answer.



- b) is carrying the 1 A current in the present of the magnetic field with magnetic density of $B = 2.5 \text{ T}$. Find the force induced on the conductor. Please write the equation and final answer.

